

Please hand in the following problems.

**NOTE:** Write answers on this page and show any work on separate pages.

**Problem 1:** Compute the following indefinite integrals. In each case, assume  $n$  is constant. Show work by applying integration techniques. (Recall :  $i = \sqrt{-1} \implies i$  is a constant)

(a)  $\int \sin(nx) dx =$

(b)  $\int e^{inx} dx =$

(c)  $\int x e^{i\pi n x} dx =$

(d)  $\int x^2 e^{inx} dx =$

**Problem 2:** Determine all possible values of the expressions below for  $n = 1, 2, 3, \dots$  by rewriting each expression as a piece-wise function of  $n$  or preferably as an equivalent single function of  $n$ . (Hint: Try some values of  $n$  to see a pattern)

(a)  $\sin(\pi n) =$   $\cos(\pi n) =$   $e^{i\pi n} =$

(b)  $\sin(2\pi n) =$   $\cos(2\pi n) =$   $e^{i2\pi n} =$

(c)  $\sin\left(\frac{\pi}{2}n\right) =$   $\cos\left(\frac{\pi}{2}n\right) =$   $e^{i\pi n/2} =$

**Problem 3:** Circle the correct answer in brackets [ ] and/or fill in any blanks.

(a) If  $f(x)$  is an even function then \_\_\_\_\_ (Look up the definition somewhere)

(b) If  $f(x)$  is an odd function then \_\_\_\_\_

(c) Cosine is an [ EVEN / ODD ] function, therefore  $\cos(-x) =$

(d) Sine is an [ EVEN / ODD ] function, therefore  $\sin(-x) =$

(e) By Euler's formula and (c) & (d),  $e^{inx} =$

(f) By Euler's formula and (c) & (d),  $e^{-inx} =$

(g) By (c)-(f),  $e^{-inx} + e^{inx} =$

(h) By (c)-(f),  $(e^{-inx} - e^{inx})i =$